

is the capability to redefine the appearance of visual controls with programmatic skins. In past versions of the Flex SDK, you could create your own skin as an ActionScript class. By overriding certain methods that are called at runtime from the Flex framework and using the Flash drawing application programming interface (API) to declare vector graphics, you could define an entire visual presentation without ever using graphical applications like Adobe Photoshop and Illustrator.

This strategy, however, was slow and cumbersome. And, because the graphical applications mentioned earlier couldn't interpret the ActionScript code and show you a preview of its results, you had to use your imagination (and a lot of graph paper) to figure out how to code the image you wanted to create. In Flex 4.5, skins are now defined as MXML components that are extended from new classes named `Skin` and `SparkSkin`. You can bind the skin component to an application or custom component at compile time or runtime with its `skinClass` style. The skin class can be constructed with a combination of vector graphics, bitmap graphics, and Flex components to achieve whatever design ideas your application requires. We will see how to define custom skins using the new MXML architecture, and then how to attach them to your applications and custom components.

Creating and Using Spark Custom Skins

A custom skin component for a Flex 4.5 application or any of its Spark components is based on the new `Skin` and `SparkSkin` classes. You can start either by creating a new copy of a component's default skin class (a step made very easy by Flash Builder 4.5) or you can create a new fresh component that extends the `Skin` class. While it's possible to create such components in either MXML or ActionScript, most developers find that the MXML approach is much easier and the resulting code far more readable.

You can create a new custom skin using the new architecture for any component that includes the new `SkinnableComponent` class in its inheritance hierarchy. `SkinnableComponent` extends `UIComponent`, which means that Spark components can be added as display children of any Spark or MX containers. In this section, we will look at how to create a new custom skin for the `Spark Application` component, but most of the techniques also apply to all other Spark components.

To create a new custom skin, start by creating a new custom MXML component. As with all MXML components and ActionScript classes, it is recommended that you place your custom skins in a specially named